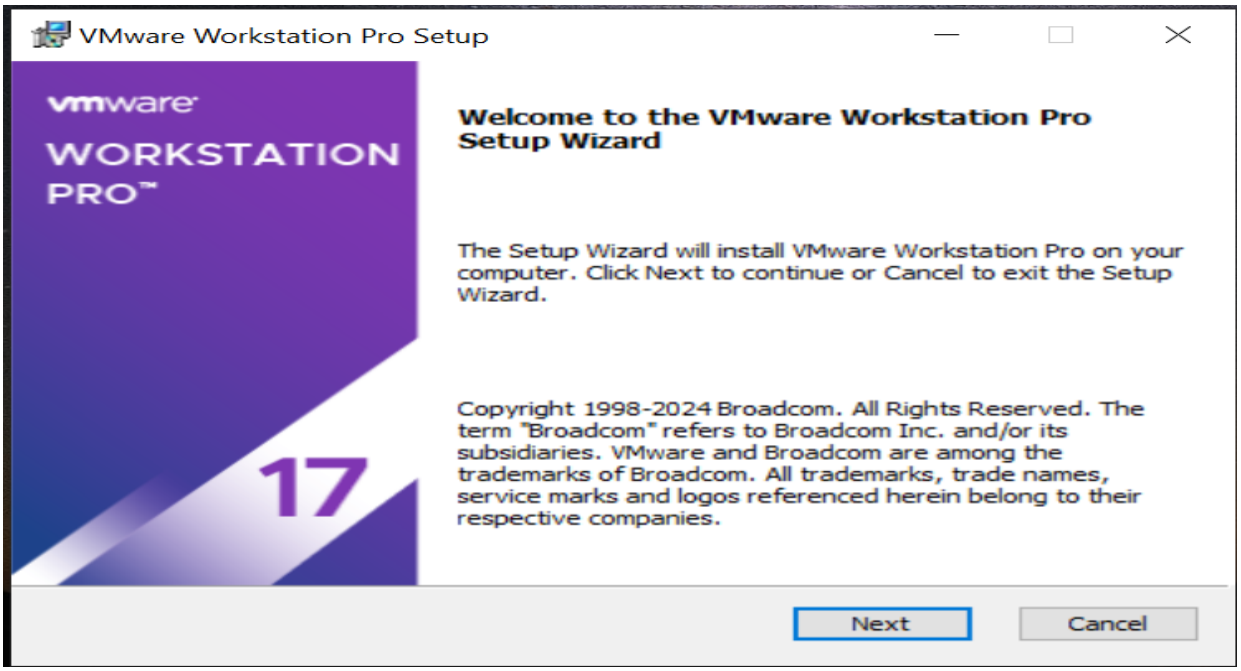
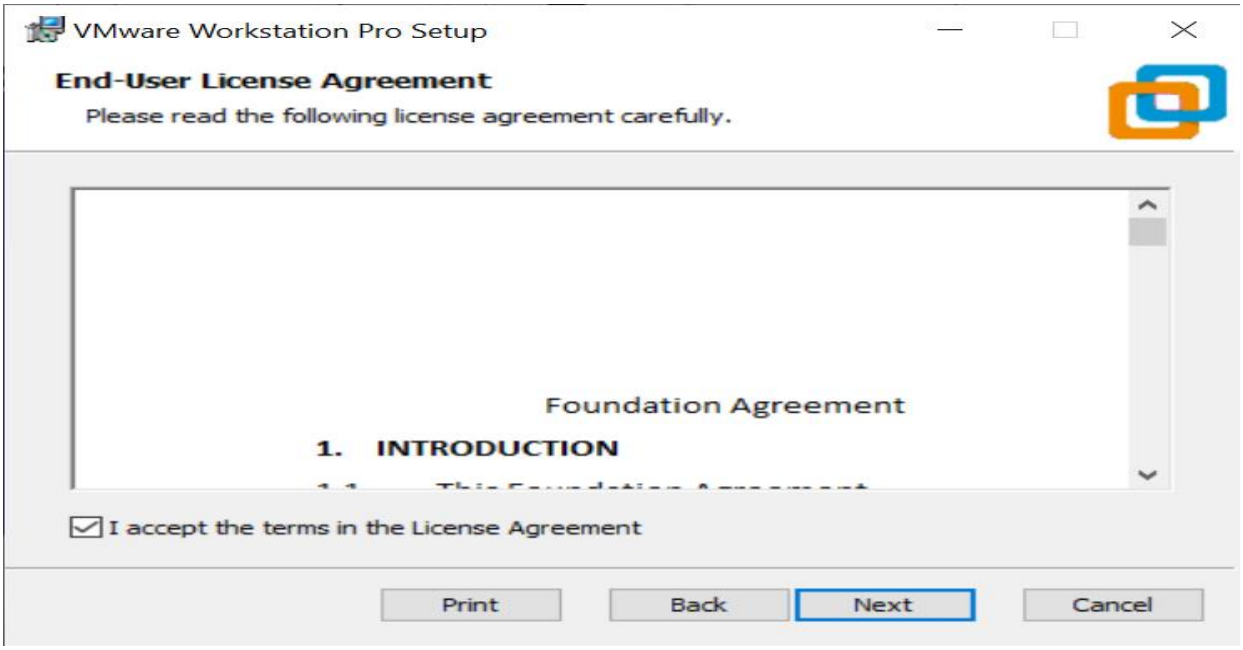
LAB 1

NAME: Amina Noor
REG NO: 2023-BSE-007
CLASS: BSE-5A
SUBJECT: CC

TASK:

1. **Install VMware Workstation Pro.**





VMware Workstation Pro Setup

Custom Setup

Select the installation destination and any additional features.

Install to:
C:\Program Files (x86)\VMware\VMware Workstation\

Change...

☑ Add VMware Workstation console tools into system PATH

Back | Next | Cancel

VMware Workstation Pro Setup

User Experience Settings

Edit default settings that can improve your user experience.

☑ Check for product updates on startup
When VMware Workstation Pro starts, check for new versions of the application and installed software components.

☑ Join the VMware Customer Experience Improvement Program

VMware's Customer Experience Improvement Program ("**CEIP**") provides VMware with information that enables VMware to improve its products and services, to fix problems, and to advise you on how best to deploy and use our products. As part of the CEIP, VMware collects technical

Learn More

Back | Next | Cancel

VMware Workstation Pro Setup

Shortcuts

Select the shortcuts you wish to place on your system.

Create shortcuts for VMware Workstation Pro in the following places:

☑ Desktop

☑ Start Menu Programs Folder

Back | Next | Cancel

VMware Workstation Pro Setup

Ready to install VMware Workstation Pro

Click Install to begin the installation. Click Back to review or change any of your installation settings. Click Cancel to exit the wizard.

Back | Install | Cancel

VMware Workstation Pro Setup

Installing VMware Workstation Pro

Please wait while the Setup Wizard installs VMware Workstation Pro.

Status:

Back | Next | Cancel

VMware Workstation Pro Setup

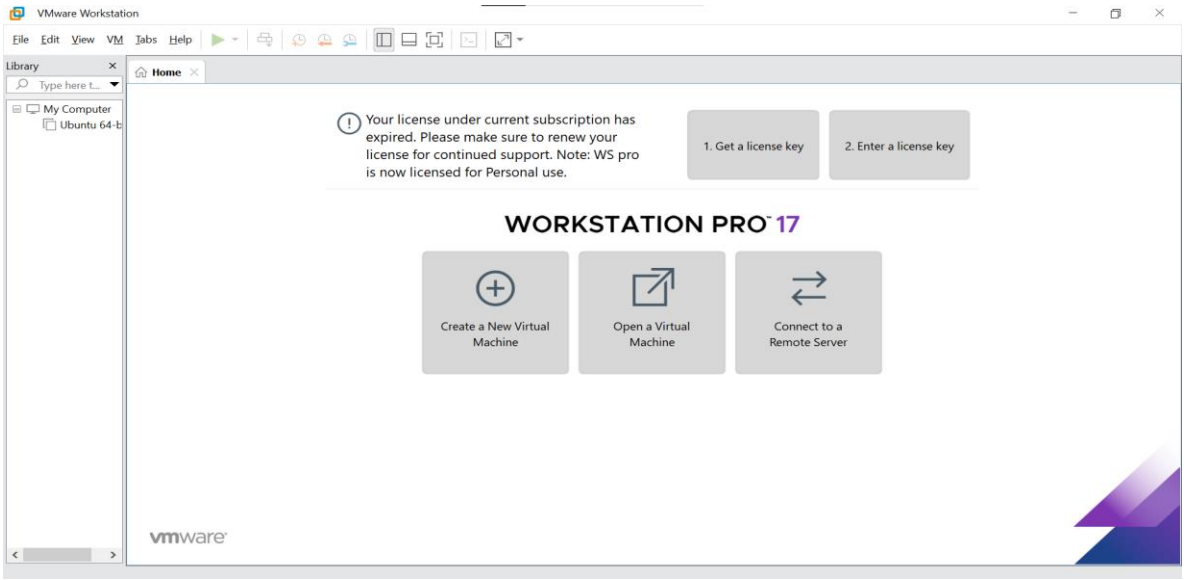
vmware
WORKSTATION PRO™
17

Completed the VMware Workstation Pro Setup Wizard

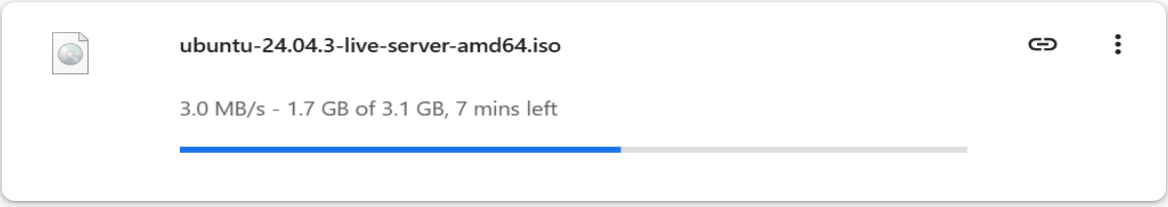
Click the Finish button to exit the Setup Wizard.

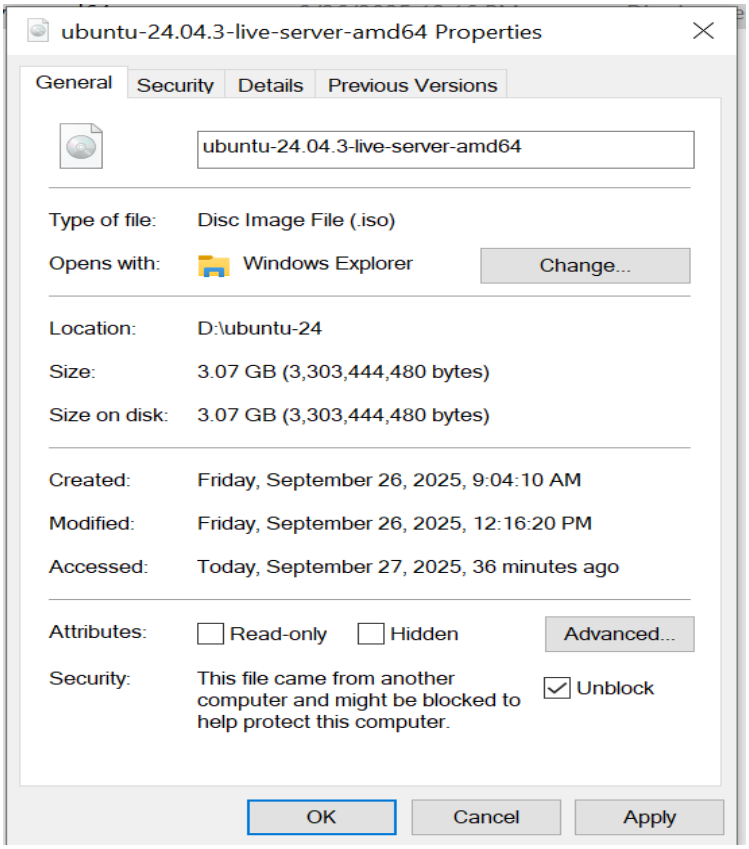
Press the License button below if you want to enter a license key now.

License | Finish



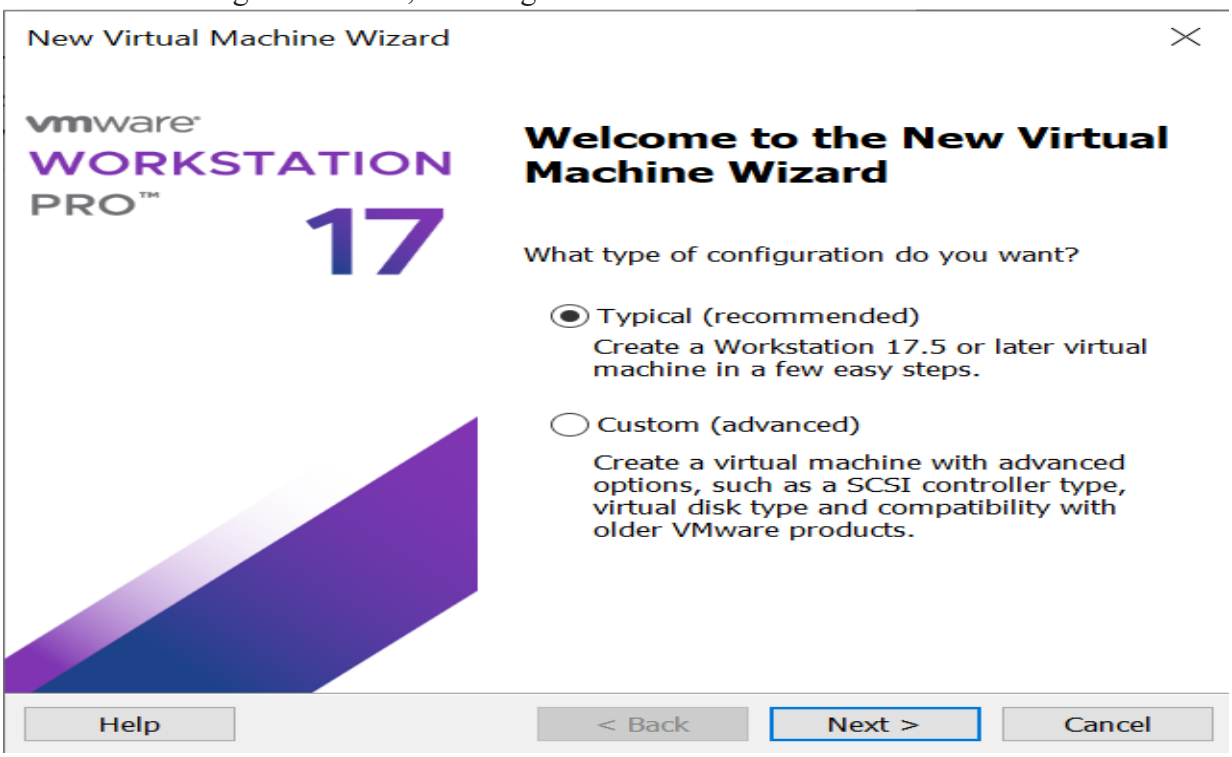
2. Download Ubuntu Server ISO

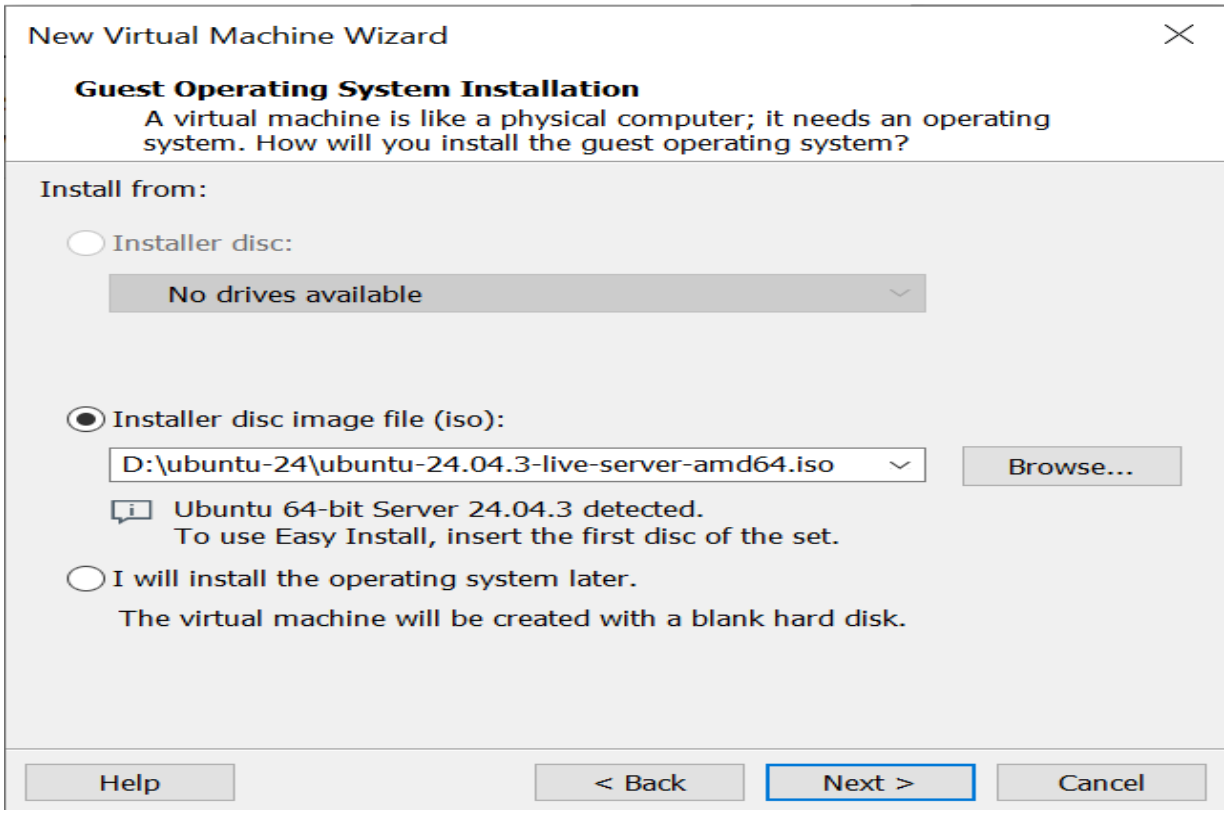




3. Create a new virtual machine in VMware

- Open VMware Workstation Pro.
- Click **Create a New Virtual Machine**.
- Select **Typical (recommended)**.
- When asked for installation media, choose **Installer disc image file (ISO)** and browse to the Ubuntu Server ISO you downloaded.
- Continue through the wizard, selecting the defaults unless otherwise instructed.





New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:
Ubuntu 64-bit (6)

Location:
C:\Users\HP\OneDrive\Documents\Virtual Machines\Ubuntu 64-bi
Browse...

The default location can be changed at Edit > Preferences.

< Back | Next > | Cancel

New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB): 25.0

Recommended size for Ubuntu 64-bit: 20 GB

- (●) Store virtual disk as a single file
- ( ) Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help | < Back | Next > | Cancel

New Virtual Machine Wizard

Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Ubuntu 64-bit.

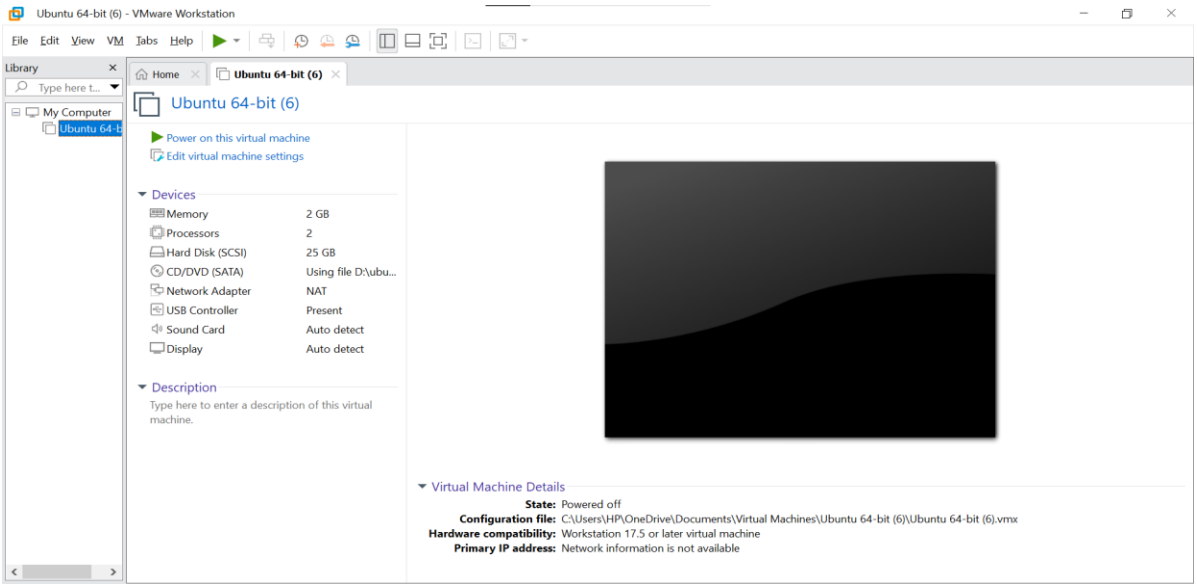
The virtual machine will be created with the following settings:

Name:	Ubuntu 64-bit (6)
Location:	C:\Users\HP\OneDrive\Documents\Virtual Machines\Ubunt...
Version:	Workstation 17.5 or later
Operating System:	Ubuntu 64-bit
Hard Disk:	25 GB
Memory:	2048 MB
Network Adapter:	NAT
Other Devices:	2 CPU cores, CD/DVD, USB Controller, Sound Card

Customize Hardware...

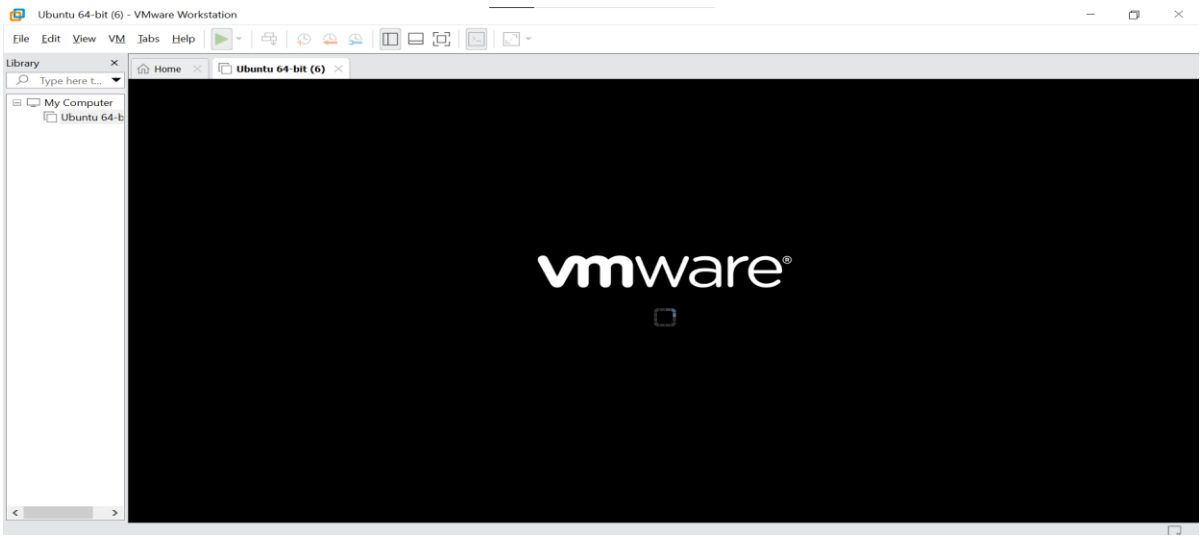
☑ Power on this virtual machine after creation

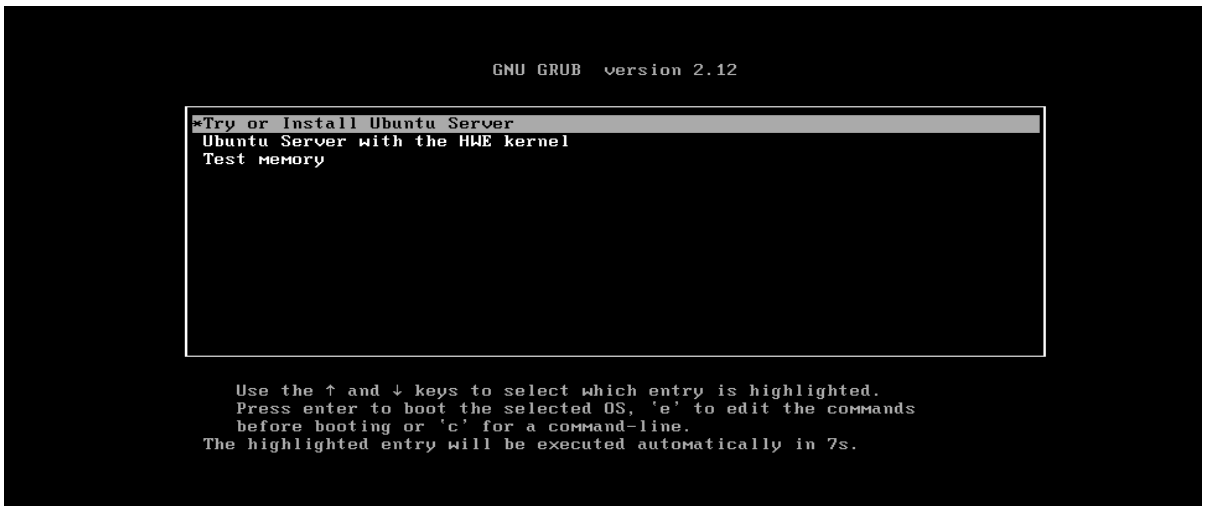
< Back | Finish | Cancel

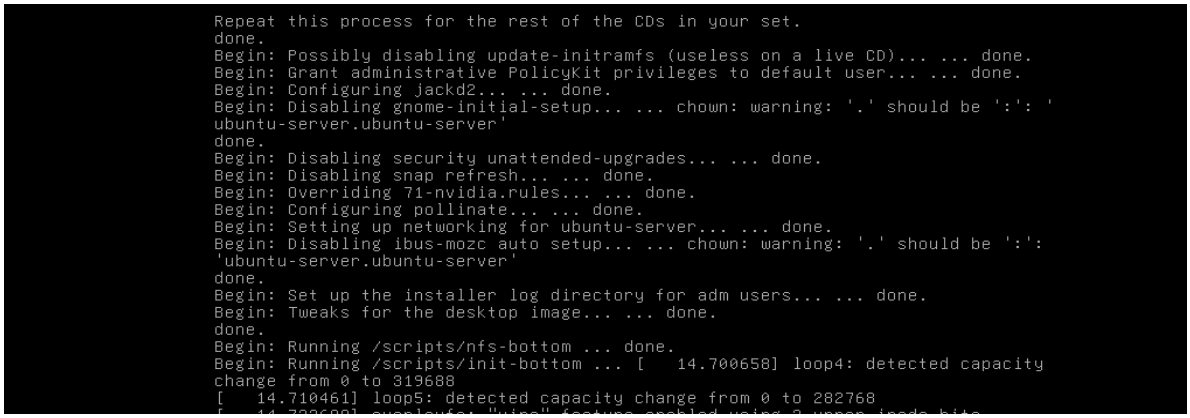


4. Start the virtual machine

- Once the VM is created, click Power on this virtual machine.
- The Ubuntu Server installer will boot using the ISO file.

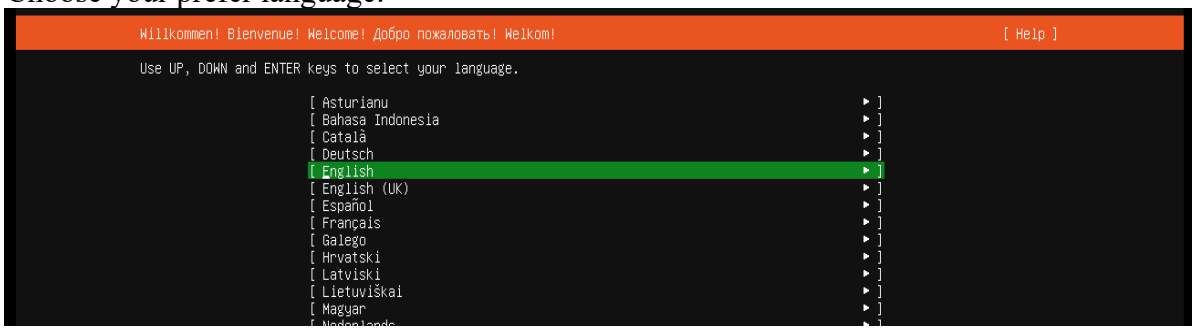




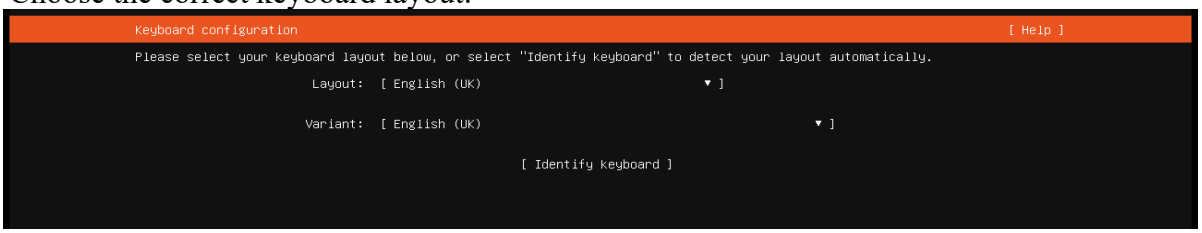


Scroll down using the arrow keys. When done, highlight **Done** at the bottom of the window and press Enter.

Choose your prefer language:

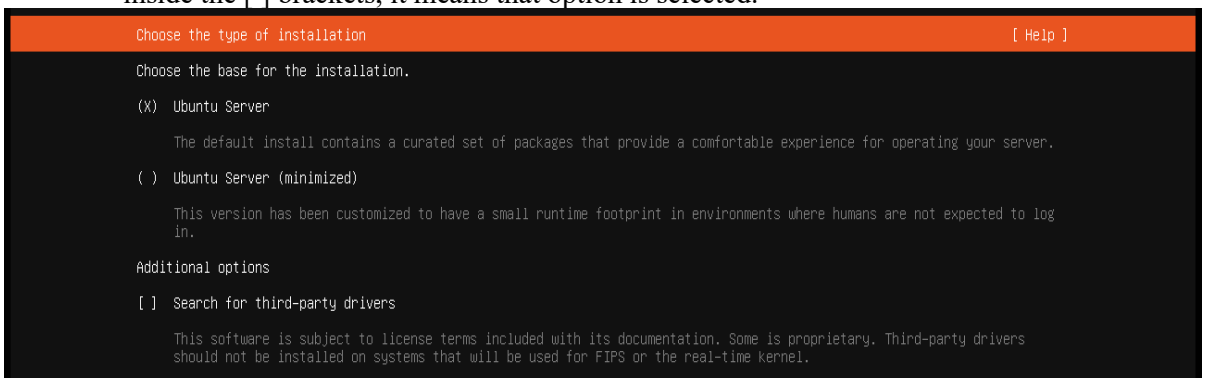


Choose the correct keyboard layout:

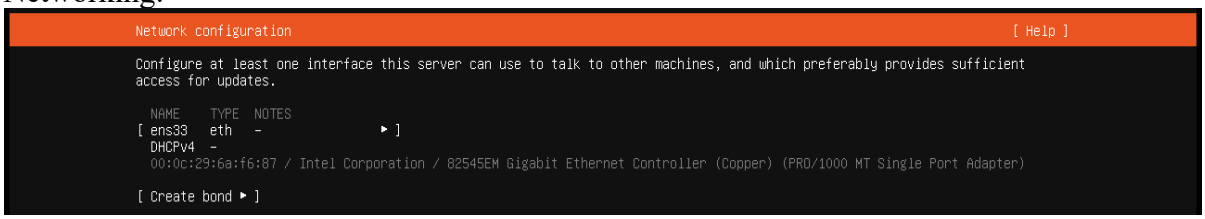


Choose your install:

- When choosing, scroll down using the arrow keys and select with the spacebar. When an (x) appears inside the [ ] brackets, it means that option is selected.



Networking:



Configure proxy:

```
Proxy configuration                                                    [ Help ]

If this system requires a proxy to connect to the internet, enter its details here.

Proxy address:
                If you need to use a HTTP proxy to access the outside world, enter the proxy information here. Otherwise,
                leave this blank.

                The proxy information should be given in the standard form of "http://[[user][:pass]@]host[:port]/".
```

Configure ubuntu archive mirror:

```
Ubuntu archive mirror configuration                                    [ Help ]

If you use an alternative mirror for Ubuntu, enter its details here.

Mirror address:  http://pk.archive.ubuntu.com/ubuntu/
                 You may provide an archive mirror to be used instead of the default.

This mirror location passed tests.

Hit:1 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists...
```

Configure storage guide:

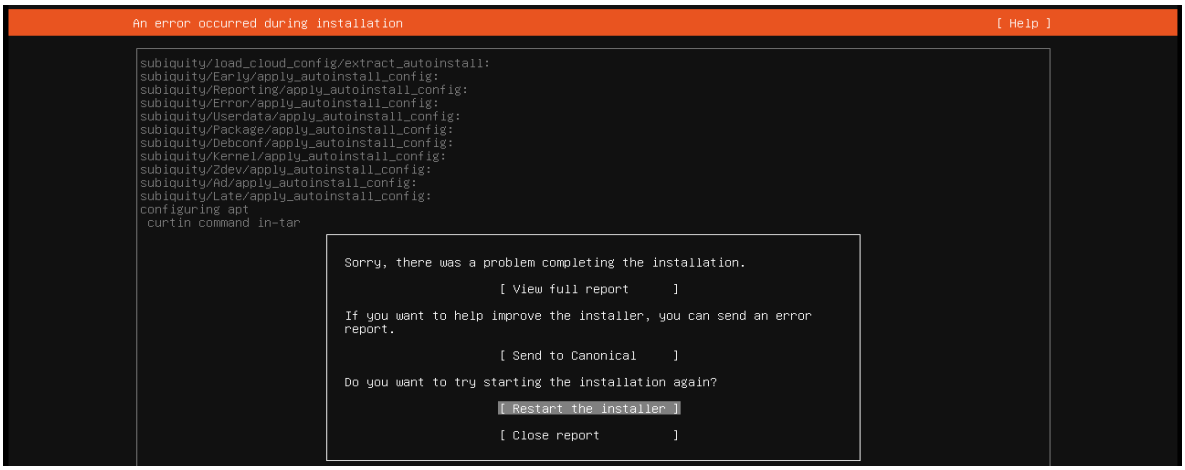
```
Guided storage configuration                                           [ Help ]

Configure a guided storage layout, or create a custom one:

(X)  Use an entire disk

     [ /dev/sda local disk 25.000G ▼ ]

     [X]  Set up this disk as an LVM group

          [ ]  Encrypt the LVM group with LUKS

                    Passphrase:

               Confirm passphrase:

                              [ ]  Also create a recovery key
                                   The key will be stored as ~/recovery-key.txt in the live system and will be copied to
                                   /var/log/installer/ in the target system.

( )  Custom storage layout
```

Storage Configuration:

```
Storage configuration                                                  [ Help ]

FILE SYSTEM SUMMARY

  MOUNT POINT     SIZE    TYPE      DEVICE TYPE
[ /              11.496G  new ext4  new LVM Logical Volume      ► ]
[ /boot           2.000G  new ext4  new partition of local disk ► ]


AVAILABLE DEVICES

  DEVICE                                    TYPE                 SIZE
[ ubuntu-vg (new)                           LVM volume group    22.996G  ► ]
  free space                                                     11.500G  ►

[ Create software RAID (md) ► ]
[ Create volume group (LVM) ► ]


USED DEVICES

  DEVICE                                    TYPE                 SIZE
[ ubuntu-vg (new)                           LVM volume group    22.996G  ► ]
  ubuntu-lv    new, to be formatted as ext4, mounted at /       11.496G  ►

[ /dev/sda                                  local disk          25.000G  ► ]
  partition 1  new, BIOS grub spacer                               1.000M  ►
  partition 2  new, to be formatted as ext4, mounted at /boot      2.000G  ►
  partition 3  new, PV of LVM volume group ubuntu-vg              22.997G  ►
```

```
                    Confirm destructive action

Selecting Continue below will begin the installation process and
result in the loss of data on the disks selected to be formatted.

You will not be able to return to this or a previous screen once the
installation has started.

Are you sure you want to continue?

                                  [ No          ]
                                  [ Continue    ]
```

Set up a profile:
Error



This usually means the installer (subiquity with curtin) had trouble during the **APT package configuration step**. Common causes are:

1. You did not check the '**Unblock**' security option in the properties of the Ubuntu Server file.
2. **No internet connection** (APT cannot fetch required packages).
3. **Bad mirror or repo unreachable** (network timeout or blocked domain).
4. **Corrupted ISO** (installer cannot unpack packages).
5. **Insufficient disk space or partitioning issue**.

Virtual machine misconfiguration (if using VMware, VirtualBox, Hyper-V, etc. – sometimes storage controllers cause problems).

Check these problems one by one, and after solving each problem, start again from Step 4 (Start the virtual machine).

Set up a profile:

Upgrade to ubuntu Pro:

SSH Configuration:

```
SSH configuration                                                    [ Help ]

You can choose to install the OpenSSH server package to enable secure remote access to your server.

[X]  Install OpenSSH server

[X]  Allow password authentication over SSH

[ Import SSH key ► ]

AUTHORIZED KEYS

  No authorized key

                          Import SSH key
Import SSH identity:  [ from GitHub    ▼ ]
                      You can import your SSH keys from GitHub or
                      Launchpad.

   GitHub Username:   hajrasarwar11
                      Enter your GitHub username.

                          [ Done      ]
                          [ Cancel    ]
```

```
SSH configuration                                                    [ Help ]

You can choose to install the OpenSSH server package to enable secure remote access to your server.

[X]  Install OpenSSH server

[X]  Allow password authentication over SSH

[ Import SSH key ► ]

AUTHORIZED KEYS

  No authorized key
```

Featured Server snaps:

```
Featured server snaps                                                [ Help ]

These are popular snaps in server environments. Select or deselect with SPACE, press ENTER to see more details of the package,
publisher and versions available.

[*] microk8s            canonical✓          Kubernetes for workstations and appliances                                  ►
[*] nextcloud           nextcloud✓          Nextcloud Server - A safe home for all your data                            ►
[ ] wekan               xet7                Open-Source Kanban                                                          ►
[*] kata-containers     katacontainers✓     Build lightweight VMs that seamlessly plug into the containers ecosystem    ►
[*] docker              canonical✓          Docker container runtime                                                    ►
[ ] canonical-livepatch canonical✓          Canonical Livepatch Client                                                  ►
[ ] rocketchat-server   rocketchat✓         Rocket.Chat server                                                          ►
[*] mosquitto           mosquitto✓          Eclipse Mosquitto MQTT broker                                               ►
[ ] etcd                canonical✓          Resilient key-value store by CoreOS                                         ►
[*] powershell          canonical✓          PowerShell for every system!                                                ►
[ ] sabnzbd             safihre             SABnzbd                                                                     ►
[*] wormhole            snapcrafters✓       get things from one computer to another, safely                             ►
[*] aws-cli             aws✓                Universal Command Line Interface for Amazon Web Services                    ►
[*] google-cloud-sdk    google-cloud-sdk✓   Google Cloud SDK                                                            ►
[*] slcli               softlayer           Python based SoftLayer API Tool.                                            ►
[*] doctl               digitalocean✓       The official DigitalOcean command line interface                            ►
[ ] postgresql10        cmd✓                PostgreSQL is a powerful, open source object-relational database system.    ►
[*] keepalived          keepalived-project✓ High availability VRRP/BFD and load-balancing for Linux                     ►
[ ] prometheus          canonical✓          The Prometheus monitoring system and time series database                   ►
[*] lxd                 canonical✓          LXD - container and VM manager                                              ►
```

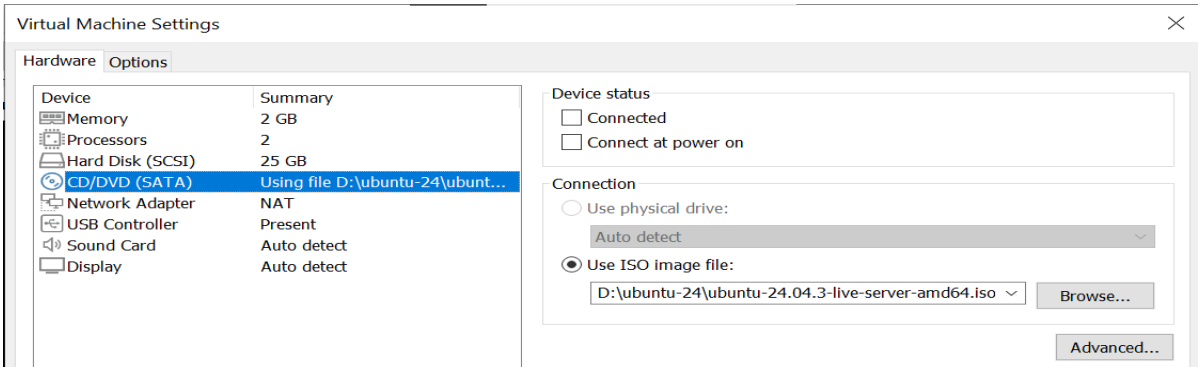
Install Software:

```
Installing system                                                    [ Help ]

subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
 curtin command in-target
installing system
 executing curtin install initial step
 executing curtin install partitioning step
  curtin command install
   configuring storage
    running 'curtin block-meta simple'
     curtin command block-meta
      removing previous storage devices
      configuring disk: disk-sda
     configuring partition: partition-0
     configuring partition: partition-1
     configuring format: format-0
     configuring partition: partition-2
     configuring lvm_volgroup: lvm_volgroup-0
     configuring lvm_partition: lvm_partition-0
     configuring format: format-1
     configuring mount: mount-1
     configuring mount: mount-0
 executing curtin install extract step
  curtin command install
   writing install sources to disk
    running 'curtin extract'
     curtin command extract
      acquiring and extracting image from cp:///tmp/tmp7gqf22di/mount
 configuring keyboard
  curtin command in-target
 executing curtin install curthooks step
  curtin command install
   configuring installed system
    running 'curtin curthooks'
     curtin command curthooks
      configuring apt configuring apt
      installing missing packages
      Installing packages on target system: ['grub-pc']
      configuring iscsi service
      configuring raid (mdadm) service
      configuring NVMe over TCP
      installing kernel |

                                   [ View full log ]
```

Error:

```
[FAILED] Failed unmounting cdrom.mount - /cdrom.
Please remove the installation medium, then press ENTER:
[FAILED] Failed unmounting cdrom.mount - /cdrom.
```

Go to Virtual Machine setting, click CD/DVD… → uncheck Connected and connect at power on Boxes → OK.

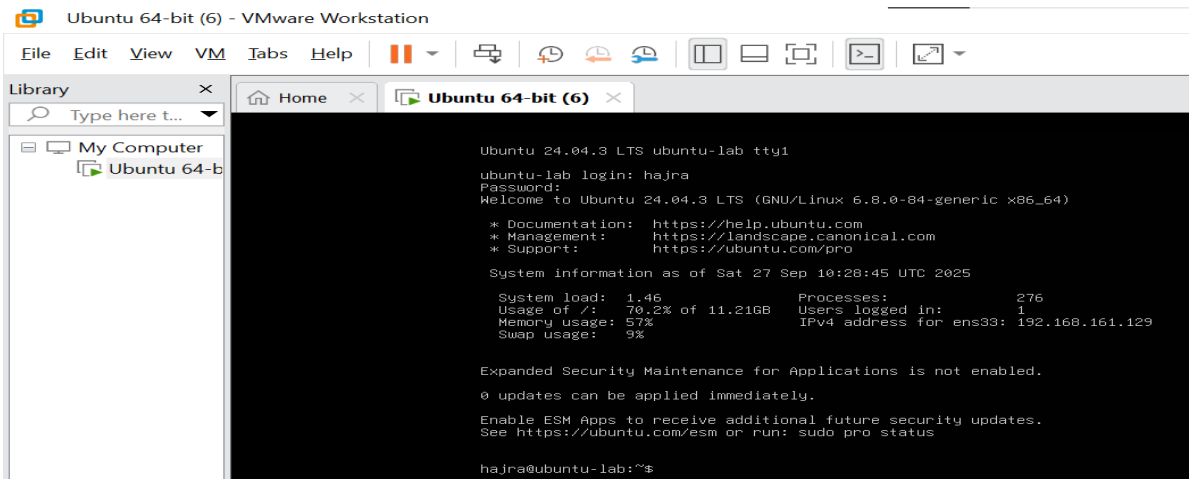



```
2236412k
[    6.505289] systemd[1]: Starting multipathd.service - Device-Mapper Multipath
 Device Controller...
         Starting multipathd.service - Devi…pper Multipath Device Controller...
[    6.507272] systemd[1]: systemd-hwdb-update.service - Rebuild Hardware Databa
se was skipped because no trigger condition checks were met.
[    6.507661] systemd[1]: systemd-pstore.service - Platform Persistent Storage
Archival was skipped because of an unmet condition check (ConditionDirectoryNotE
mpty=/sys/fs/pstore).
[    6.556528] systemd[1]: Starting systemd-random-seed.service - Load/Save OS R
andom Seed...
         Starting systemd-random-seed.service - Load/Save OS Random Seed...
[    6.558384] systemd[1]: systemd-repart.service - Repartition Root Disk was sk
ipped because no trigger condition checks were met.
[    6.580600] systemd[1]: Starting systemd-sysctl.service - Apply Kernel Variab
les...
         Starting systemd-sysctl.service - Apply Kernel Variables...
[    6.607339] systemd[1]: Starting systemd-tmpfiles-setup-dev-early.service - C
reate Static Device Nodes in /dev gracefully...
         Starting systemd-tmpfiles-setup-de… Device Nodes in /dev gracefully...
[    6.608038] systemd[1]: systemd-tpm2-setup.service - TPM2 SRK Setup was skipp
ed because of an unmet condition check (ConditionSecurity=measured-uki).
[    6.608585] systemd[1]: Activated swap swap.img.swap - /swap.img.
[  OK  ] Activated swap swap.img.swap - /swap.img.
```

```
Ubuntu 24.04.3 LTS ubuntu-lab tty1

ubuntu-lab login: _
```

```
Ubuntu 24.04.3 LTS ubuntu-lab tty1

ubuntu-lab login:          Mounting snap-snapd-25202.mount - Mount unit for snapd, revision 25202...
[  OK  ] Mounted snap-snapd-25202.mount - Mount unit for snapd, revision 25202.
         Starting snapd.service - Snap Daemon...
[  OK  ] Started snapd.service - Snap Daemon.

ubuntu-lab login: hajra
Password: _
```

```
[  OK  ] Created slice user-1000.slice - User Slice of UID 1000.
         Starting user-runtime-dir@1000.service - User Runtime Directory /run/user/1000...
[  OK  ] Finished user-runtime-dir@1000.service - User Runtime Directory /run/user/1000.
         Starting user@1000.service - User Manager for UID 1000...
[  OK  ] Started user@1000.service - User Manager for UID 1000.
[  OK  ] Started session-1.scope - Session 1 of User hajra.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

hajra@ubuntu-lab:~$          Mounting snap-core22-2133.mount - Mount unit for core22, revision 2133...
[  OK  ] Mounted snap-core22-2133.mount - Mount unit for core22, revision 2133.

hajra@ubuntu-lab:~$          Mounting snap-aws\x2dcli-1597.mount - Mount unit for aws-cli, revision 1597...
[  OK  ] Mounted snap-aws\x2dcli-1597.mount - Mount unit for aws-cli, revision 1597.

hajra@ubuntu-lab:~$          Mounting snap-docker-3265.mount - Mount unit for docker, revision 3265...
[  OK  ] Mounted snap-docker-3265.mount - Mount unit for docker, revision 3265.
[  OK  ] Started snap.docker.hook.install-bb1f3125-37b5-4ed2-8d3a-6a6c117a275a.scope.
         Starting snap.docker.nvidia-container-toolkit.service - Service for snap application docker.nvidia-container-toolkit...
[  OK  ] Finished snap.docker.nvidia-container-toolkit.service - Service for snap application docker.nvidia-container-toolkit.
[  OK  ] Started snap.docker.dockerd.service - Service for snap application docker.dockerd.
[  OK  ] Started snap.docker.hook.configure-c5604d03-6da2-4743-9716-314a124a5a94.scope.
         Starting snap.docker.nvidia-container-toolkit.service - Service for snap application docker.nvidia-container-toolkit...
[  OK  ] Finished snap.docker.nvidia-container-toolkit.service - Service for snap application docker.nvidia-container-toolkit.
         Mounting snap-doctl-2242.mount - Mount unit for doctl, revision 2242...
[  OK  ] Mounted snap-doctl-2242.mount - Mount unit for doctl, revision 2242.
         Starting update-notifier-download.service - Download data for packages that failed at package install time...
[  OK  ] Finished update-notifier-download.service - Download data for packages that failed at package install time.
_
```



Ubuntu Server Installed Successfully.

5. Accessing your Ubuntu Server from Windows

1. Find the IP address of your Ubuntu Server using "ip addr"

```
ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
  link/ether 00:0c:29:6a:f6:87 brd ff:ff:ff:ff:ff:ff
  altname enp2s1
  inet 192.168.161.129/24 metric 100 brd 192.168.161.255 scope global dynamic ens33
     valid_lft 1082sec preferred_lft 1082sec
  inet6 fe80::20c:29ff:fe6a:f687/64 scope link
     valid_lft forever preferred_lft forever
```

2. Connect via SSH from Windows

```
PS C:\Users\HP> ping 192.168.161.129

Pinging 192.168.161.129 with 32 bytes of data:
Reply from 192.168.161.129: bytes=32 time=1ms TTL=64
Reply from 192.168.161.129: bytes=32 time<1ms TTL=64
Reply from 192.168.161.129: bytes=32 time<1ms TTL=64
Reply from 192.168.161.129: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.161.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

3. **Accept the fingerprint** (first time only)
   Type yes when prompted.

```
PS C:\Users\HP> ssh hajra@192.168.161.129
The authenticity of host '192.168.161.129 (192.168.161.129)' can't be established.
ED25519 key fingerprint is SHA256:M5unwK7+jxUhr6KM3bbMPRkghlGd+Kq5uspTQZ5Up3w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

4. **Enter your password**
   Use the same password you set up during the Ubuntu Server installation.

```
Warning: Permanently added '192.168.161.129' (ED25519) to the list of known hosts.
hajra@192.168.161.129's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-84-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat 27 Sep 10:28:45 UTC 2025

  System load:  1.46                Processes:              276
  Usage of /:   70.2% of 11.21GB    Users logged in:        1
  Memory usage: 57%                 IPv4 address for ens33: 192.168.161.129
  Swap usage:   9%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


hajra@ubuntu-lab:~$
```